# Compass: final results and parameters

Research summary as of 2026-10-22. Every number is from a committed, pre-registered study in the georgeredo repository (branch `claude/lucid-heisenberg-bpyq7k`); commit hashes are given so each can be checked.

**Bottom line.**

- No strategy tested is ready to trade with real money. None has passed its pre-registered test and its forward confirmation.
- Buying calls and puts loses after costs: about −13% of the debit per trade on the 0DTE, index and ETF base trades, and about −23% on 31–60 DTE single-stock options.
- No direction signal tested beats that cost: UW flow and sweeps, net premium, NOPE, market tide, dark pool, GEX regime, walls and flip, opening range, prior-day levels, and post-earnings drift.
- Three rules reduce the loss and passed a held-out test: a **spread veto**, a **SPXW-vs-SPY venue choice** and a **14:00 exit on 0DTE**. Together they cut the loss from about −11% to about −4% per trade. They don't make long options profitable.
- The short-premium paper candidates (iron fly, 20-delta condor) looked positive at the official close, but fail once after-hours assignment risk is priced.

## 1. Costs and fills used everywhere

| Item | Value |
|---|---|
| Fills | Buy at the ask, sell at the bid (NBBO, 1-minute data from ThetaData). No mid fills assumed. |
| Fees, SPY / QQQ / IWM / single stocks | $0.68748 per contract each way ($0.65 commission + Cboe ORF $0.01248 + OCC $0.025). None at expiry. |
| Fees, SPXW | $1.18748 (premium < $1) / $1.27748 (≥ $1) per contract each way: commission + Cboe SPX/SPXW customer fee + SPXW surcharge + ORF + OCC (cited in PREREG_SPXW.md). |
| Settlement | Official closing price (UW daily bar; Theta EOD before 2025-10-13; Cboe's official SPX close for SPXW). Physically settled names are never held past 15:58 in the feature study. |
| Statistics | Session sums; 20-session block bootstrap; periods never pooled; a pass needs the mean above zero with the interval clear of zero. |

## 2. Results, study by study

### 2.1 Paper tracker candidates (short premium, SPY/QQQ)

Period ii (2025-01-02 → 2026-10-09). Mean $ per trade at the official close, then with our short legs assigned on the 17:30 after-hours price ("assignment-risk case"), with 20-session block intervals.

| Candidate | Official close | Assignment-risk case [95%] | 15:58 exit | Verdict |
|---|---|---|---|---|
| fly5 iron (SPY/QQQ ±5 iron fly, 15:00/15:30) | +$16.20 to +$18.25 | +$22.52 [−20.07, +52.89] | −$13.13 | **Drop** |
| condor_d20 (SPY 2-wide 20-delta condor, 13:00–15:00) | +$7.10 to +$7.67 | +$3.84 [−19.94, +22.74] | −$6.29 | **Drop** |
| condor_em1 (SPY midday condor outside the straddle) | +$4.10 [−0.51, +8.15] | — | negative | **Reject** |
| fly5 call / put flies | +$7.13 / +$5.58 | — | negative | **Reject (cash account)** |

The official-close column is two figures: the audit's (to 2026-10-09) and the after-hours study's (period ii's overlap). The after-hours study is `633fbff`. Period iii (7 forward sessions) is too short for an interval. The 2025-04-09 SPY official close of 548.62 could not be confirmed: second sources put it at $543.0–543.7.

### 2.2 SPXW versions of the trades, pre-registered (`2820bb4` / `6bb230e`)

| Structure | Development | Selection [95%] | Verdict |
|---|---|---|---|
| Iron fly 15:00/15:30 | +$79.78 / session | −$39.64 [−173, +87] | **Fail** |
| 20-delta condor 13:00–15:00 | +$35.29 | +$120.76 [−95, +287] | **Fail** |

SPY and QQQ options carry more late-day premium than SPXW relative to the realized move. At 15:58 the gap is +0.80 (SPY) and +1.39 (QQQ) in the selection period, and every interval clears zero. That fits an assignment premium, but collecting it means carrying the assignment.

### 2.3 The 0DTE long call/put rule from the parameter writeup (`c502122` / `1e10053`)

Opening-range sweep fade 10:00–11:59; nearest out-of-the-money 0DTE strike at the ask; stop 0.68 × ask; target 1.25 × ask; 5-minute rule; flat 15:58. The 134 dates the cards came from were excluded.

| Name | Trades | Mean per trade [95%] | Win rate | Verdict |
|---|---|---|---|---|
| SPY | 777 | −6.28% [−8.18, −4.41] | 34.5% | **Fail** |
| QQQ | 820 | −5.21% [−7.10, −3.56] | 36.6% | **Fail** |
| IWM | 438 | −9.47% [−12.11, −7.18] | 34.9% | **Fail** |
| SPXW | 908 | −4.96% [−6.68, −3.49] | 37.0% | **Fail** |

The narrative exits (breakeven lock, VIX trail), a hard 5-minute exit and the "improved" parameters all fail the same way. Grade check (descriptive, GeorgeFi's own ledger): A+ +8.1%, A +3.1%, B+ −3.2% per trade. A+ beats B+ with an interval clear of zero, but A+ does not separate from A.

### 2.4 Which features improve which decision, pre-registered (`a2e14e4` → `2a49049`, `b6737c7` → `e5dd8c6`)

Universe: SPY, QQQ, IWM, SPXW and 11 large stocks. Base trades: long calls and puts, 0DTE every 30 minutes and 1–60 DTE at the close. Holdout: AVGO, MU and QQQ plus 43 of 173 months.

- **GEX check: failed.** Negative-GEX days did not move more than positive-GEX days (−0.06 implied moves [−0.13, +0.04]), so the GEX regime, flip and wall features were dropped untested.

- 23 features screened, 3 passed the screen, 3 passed the holdout.

| Feature | Decision | Exploration gain [95%] | Holdout gain [95%] | Verdict |
|---|---|---|---|---|
| V1 spread veto (10% of mid) | Trade / no trade | +2.46 [+1.64, +3.41] | +2.61 [+1.76, +3.46] | **Pass** |
| VN1 cheaper of SPXW and SPY | Venue | +1.69 [+0.99, +2.26] | +1.18 [+0.50, +2.20] | **Pass*** |
| X3 sell 0DTE at 14:00 | Exit | +5.75 [+1.83, +9.44] | +7.78 [+4.14, +10.95] | **Pass** |
| Other 17 (OI, volume, earnings, macro, IV rank, term structure, implied vs realized, strike rule, delta, implied-move target, all 7 direction features) | Various | Failed the screen | | **Fail** |

Gains are percentage points of the debit per trade. *The holdout was read twice after a duplicated-row bug, all on held-out days, was fixed. VN1 failed the first read and passed only the corrected one, so it is the weakest of the three. Improvement round: 33 variants tried, no direction variant survived; the three candidates below passed, but on a holdout already read for V1 and X3.

### 2.5 Post-earnings drift, pre-registered (`38b11bf` → `94fb69d`)

Call after a positive earnings-day return, put after a negative one, 31–60 DTE, entered the session after the reaction. Exploration: 3,129 events on 1,601 names, 2019–2022.

**Δ = −0.90 points [−3.53, +1.71]: Fail.** Mean per trade: −24.3% on the picked side, −22.5% on the other. The holdout (1,600 names never opened by any study) was not fetched and stays clean for one future test. The overnight-reversal hypothesis does not exist as a study, so no options version was added.

## 3. Final parameters (frozen)

### 3.1 Loss-reduction models in forward confirmation

| Model | Exact rule | Holdout gain | Forward sessions needed (N) |
|---|---|---|---|
| V1 | Skip a contract when (ask − bid) ÷ mid at entry > 10%. | +2.61 | 23,960 (never, in practice) |
| BV1b | Same veto at 3%. | +4.30 | 481 |
| VN1 | Between SPXW and SPY at the same minute and strike distance (implied-move units), keep the lower ((ask − mid) + fee ÷ 100) ÷ mid; a tie keeps both. | +1.18 | 587 |
| X3 | A 0DTE trade entered before 14:00 is sold at the 14:00 bid instead of being held. | +7.78 | 369 |
| BX3e | X3 on physically settled names only (SPY, QQQ, IWM, stocks). | +7.67 | 265 |
| BC3 | V1 veto + X3 exit + direction by the opening range: the 09:30–09:59 high/low; at entry, a close above the high → call, below the low → put, inside → no trade. | +6.77 (still −4.26% per trade) | 341 |

Rule (`b101ae1`): each model is judged once, after exactly N complete forward sessions from 2026-10-22. It is confirmed if Δ > 0 with the 20-session block interval clear of zero. N gives 80% power for an effect the size the holdout showed. The forward set is SPY, QQQ and SPXW 0DTE, so about 1–2.3 years at 252 sessions a year.

### 3.2 Base trades the models are judged against (PREREG_FEATURES.md)

- **0DTE:** entries at 09:45, 10:15 … 15:15, 15:30. Exit at the 15:58 bid; SPXW settles on the official SPX close.
- **1–7 DTE:** held to expiry. **8–30 DTE:** closing bid 5 sessions later. **31–60 DTE:** closing bid 10 sessions later. Entry at the closing ask; the expiry nearest 4, 19 or 45 days.
- **Strike buckets**, in implied-move units (ATM straddle mid ÷ spot): ITM −0.5 to −0.15, ATM ±0.15, OTM1 0.15–0.5, OTM2 0.5–1.0, OTM3 1.0–1.5.
- **Liquidity tiers**, measured on the 20 sessions before each month: Tier 1 is volume ≥ 900k and spread ≤ 5%; Tier 3 is volume < 400k or spread > 10%.

### 3.3 Paper tracker candidates (still logged daily)

| Candidate | Parameters |
|---|---|
| condor_em1 | SPY 2-wide iron condor; shorts at the first strike outside spot ± the ATM straddle; entries 11:30–13:30. |
| fly5 | SPY/QQQ ±5 iron, call and put flies at 15:00 and 15:30; centre at the strike nearest spot. |
| condor_d20 | SPY 2-wide 20-delta iron condor, 13:00–15:00; its D1 subset uses the frozen model deciles. |
| fly_spxw | SPXW ±5 / ±25 flies. |

Every tracker row carries the official-close settlement, a 15:58-exit comparison with the cost stress, flags for stock left after expiry, pin distance and an after-hours strike cross, a period label, and the after-hours P&L cases.

## 4. What is running

- The **daily forward job** runs in this container every weekday at 7:47 pm New York. It downloads the day's data, checks and merges it, replays the paper tracker, and scores the six frozen models once Cboe

publishes the official SPX close.
- Each run commits the updated cumulative report and the forward files. There is no verdict before each model's N.

## 5. Caveats

- The UW key is a trial key: trade-by-trade flow history reaches back only 30 days, so flow was tested through UW's flow alerts. UW serves no past intraday GEX book, so walls and the flip are the previous session's.
- The feature study's improvement-round passes are weaker evidence, because the holdout had already been read for their components.
- All results are research. None is investment advice or a recommendation to trade.